MYCO AI

AI-Powered Mushroom Cultivation & Predictive Monitoring System

PROJECT CONCEPT • HARDWARE ARCHITECTURE • IoT DASHBOARD • AI ROADMAP

A smart miniature mushroom-growing warehouse that combines environmental sensing, automated climate control, live IoT monitoring, and data-driven growth analysis.

Prepared as a development reference

14 October 2026

1. Executive Summary

MYCO AI is a proposed smart cultivation platform for a miniature mushroom-growing warehouse. An ESP32 collects environmental readings from a temperature/humidity sensor and a growing-medium moisture sensor. A control layer operates a cooling fan, a heating bulb through a relay, and an ultrasonic humidifier. A web application presents live readings, history, alerts, cultivation batches, and AI-assisted analysis.

The software is intended to be the primary innovation: it should connect measured conditions with grower-recorded observations, track cultivation stages, and gradually develop useful predictions as real cultivation data accumulates. The system should clearly separate sensor measurements, simulated demo values, rule-based alerts, and AI-generated estimates.

Core principle: begin with reliable monitoring and safe deterministic control; introduce predictive models only when enough trustworthy data and cultivation outcomes are available.

2. Project Objectives

- Monitor temperature, relative humidity, and growing-medium moisture.
- Automate heating, air circulation, and humidification within configured limits.
- Provide a responsive IoT dashboard with live status and historical charts.
- Record batches, species/strain, substrate, inoculation date, stage observations, and harvest outcomes.
- Use an AI assistant to explain readings and cultivation records, while keeping hardware control deterministic and safety-checked.
- Build a dataset that can later support validated stage-transition, environmental forecasting, and harvest-window models.

3. Proposed Hardware

Component	Purpose	Implementation notes
ESP32 development board	Main controller and telemetry gateway	Publishes timestamped readings and receives authenticated commands where appropriate.
DHT11	Air temperature and relative humidity	Suitable for early prototype; evaluate accuracy at sustained high humidity before relying on it for control.
Moisture sensor	Growing-medium moisture indication	Identify sensor type and calibrate for the chosen substrate; raw readings are not universal moisture percentages.
12 V brushless fan	Air circulation / ventilation	Prefer a fan over a bare motor; fan alone cannot cool below ambient temperature.
Tungsten bulb + relay	Heating	Use a properly rated, enclosed and fused mains circuit; keep mains wiring away from moisture.
USB ultrasonic humidifier	Humidity increase	Retain its driver PCB; inspect button circuit before selecting a transistor/MOSFET interface.
Power supplies and protection	Reliable low-voltage power and safety	Check voltage/current ratings, common-ground requirements, fusing, isolation, and enclosure.

Hardware details still to confirm

- Exact moisture sensor type and output voltage.
- Fan wire count, rated current, and whether PWM speed control is needed.
- Humidifier PCB supply rating and the electrical behavior of its push button.

- Relay module input requirements and the bulb's mains rating.
- Selected mushroom species and cultivation method, which determine target conditions.

4. System Architecture

The system should be modular. The web UI must remain usable in demo mode before the hardware is ready, and the same telemetry schema should be used by the simulator and the physical ESP32.

Layer	Responsibilities
Sensors and actuators	DHT11, substrate moisture sensor, fan, heater relay, humidifier.
ESP32 firmware	Read sensors, validate basic ranges, apply local safety logic, report actuator status, and publish telemetry.
Communication	MQTT for scalable device messaging; HTTPS APIs and WebSockets/SSE for dashboard interaction.
Backend	Authentication, device registry, validation, control rules, data storage, alerts, prediction service, AI provider adapter.
Database	Timestamped sensor history, batches, profiles, observations, predictions, commands, and alerts.
Frontend	Overview, live charts, cultivation batches, growth timeline, AI assistant, controls, analytics, and settings.

Illustrative telemetry payload

```
{"device_id":"myco-esp32-01", "timestamp":"ISO-8601", "temperature_c":24.5,
"relative_humidity_percent":88.0, "substrate_moisture_raw":1850, "fan_state":false,
"heater_state":false, "humidifier_state":true}
```

The values above are examples only. They are not measurements from the user's hardware. Store source and quality metadata, and show whether each reading is live, simulated, stale, or invalid.

5. Website / Dashboard Features

- **Overview:** current readings, configured ranges, device status, active alerts, and recent trends.
- **Live Monitoring:** real-time charts for temperature, relative humidity, substrate sensor output, and actuator states.
- **Cultivation Batches:** species, strain, substrate, dates, target profile, notes, images, and harvest outcomes.
- **Growth Timeline:** configurable stages, grower-confirmed milestones, expected next milestone, and uncertainty labels.
- **AI Predictions:** environmental trends, anomaly explanations, stage estimates, and harvest-window estimates only when supported by data.
- **AI Cultivation Assistant:** answers based on database readings and batch context; separates measurements from predictions.
- **Environmental Control:** target ranges, monitoring/automatic/manual modes, interlocks, and command history.
- **Analytics and Reports:** historical charts, time outside target ranges, actuator runtime, CSV export, and batch reports.
- **Device Management:** connectivity, last telemetry time, firmware/device identity, and diagnostic status.

6. Suggested Software Stack

Area	Suggested option	Why
Frontend	React + TypeScript + Vite + Tailwind CSS	Responsive, maintainable interactive dashboard.
Charts	Recharts or equivalent	Time-series visualizations and interactive tooltips.
Backend	FastAPI or Node.js/TypeScript	REST endpoints, AI proxy, control logic, and real-time transport.
Database	SQLite for prototype; PostgreSQL for deployment	Persistent records and batch history.
Device messaging	MQTT broker	Telemetry and command transport with device-level access controls.
AI integration	Server-side provider adapter	Protects API secrets and permits model/provider replacement.

7. AI and Prediction Strategy

A language model is useful for interpreting observations and explaining trends, but it should not be treated as a validated biological growth model. MYCO AI should separate three functions:

- **Deterministic control:** rule-based decisions with hysteresis, maximum runtime limits, stale-sensor handling, and device-side safety logic.
- **Predictive analytics:** statistical or machine-learning models evaluated against actual future readings and recorded cultivation outcomes.
- **AI assistant:** explains measured data, history, alerts, and model outputs; it does not directly command relays or motors.

Recommended development sequence

- Start with thresholds, moving averages, trend estimates, and anomaly rules.
- Record environmental conditions and grower-confirmed stage transitions for every batch.
- Compare predictions with actual outcomes and track prediction error.
- Train or evaluate models only when the dataset is sufficient and representative.
- Report uncertainty and data limitations; never fabricate model accuracy or confidence.

Potential future predictions

- Short-term temperature and humidity trends.
- Likely time until a manually verified cultivation milestone.
- Estimated harvest window based on species, strain, substrate, stage observations, and historical outcomes.
- Environmental anomalies and recurring periods outside the configured target range.

8. Control Logic and Safety

The ESP32 must retain local safety logic so a network outage or backend failure cannot cause uncontrolled heating. Software alone is not a substitute for a physical safety mechanism.

- Use configurable target ranges specific to the selected mushroom species and cultivation method.
- Use hysteresis to avoid rapid ON/OFF switching around a single threshold.
- On invalid or stale temperature readings, disable automatic heating and enter a defined safe state.
- Use a suitable independent temperature cutoff, fuse, insulated enclosure, strain relief, and correctly rated switching hardware.
- Keep mains-voltage bulb wiring isolated from low-voltage electronics and moisture.
- Do not expose public relay-control endpoints. Authenticate users and validate every command.
- Distinguish requested actuator state from confirmed state; a sent command does not prove that a device switched.
- Place the DHT11 away from direct mist and direct heater radiation. Avoid directing airflow straight onto the substrate.

9. AI API Key Security

The API key previously pasted into chat should be treated as exposed. Revoke or rotate it before use. Do not put secrets in React code, browser bundles, VITE_ environment variables, public repositories, or logs.

- Store the replacement key in a backend-only environment variable such as AI_API_KEY.
- Keep .env out of version control and provide a .env.example containing placeholders only.

- Verify the provider's official documentation for the correct endpoint, model name, authentication, rate limits, and supported features.
- Use server-side timeouts, input validation, usage limits, safe error handling, and response validation.
- Keep the dashboard usable when the AI service is unavailable; clearly label fallback or demo behavior.

Official API documentation should be checked during implementation because provider endpoints and model availability can change.

10. Data Model

- Devices: device ID, name, firmware version, last-seen time, connection status.
- Sensor readings: device ID, timestamp, sensor type, value, unit, quality, and source.
- Cultivation batches: species, strain, substrate, dates, profile, current stage, and outcome.
- Observations: timestamp, stage confirmation, notes, photos, and grower identity where applicable.
- Predictions: model/version, input window, output, generated time, uncertainty, and later actual outcome.
- Commands and events: requested state, result state, actor, timestamp, validation result, and error details.
- Alerts: type, severity, source, opened/acknowledged/resolved timestamps, and notes.

11. Development Roadmap

Milestone	Deliverables	Completion test
1. Dashboard foundation	Navigation, responsive layout, demo telemetry, charts, and visible demo status.	Application runs locally; no fabricated live readings.
2. Batch and timeline	Batch forms, profiles, observations, stage timeline, and persistent storage.	Create/edit batches and retrieve them after restart.
3. AI assistant	Secure backend adapter, contextual chat, provider errors, and rate limits.	Secret remains server-side; chatbot uses actual stored context.
4. Device integration	MQTT telemetry, device status, ESP32 schema, command status, and reconnect behavior.	Real telemetry appears with timestamps and source labels.
5. Safe control	Hysteresis, interlocks, stale-sensor handling, event log, and hardware safety plan.	Tests prove invalid sensor data cannot trigger unsafe heating.
6. Analytics and prediction	Historical summaries, baseline forecasts, evaluation, and uncertainty.	Prediction results are compared with real outcomes.

12. Testing Checklist

- Frontend navigation, responsive layout, form validation, charts, time filters, and CSV export.
- Sensor payload validation, timestamps, stale readings, malformed messages, and duplicate messages.
- Database persistence, migrations, and batch isolation.
- AI provider timeout, invalid key, rate limit, and unavailable service.
- Authentication, authorization, secret handling, and control-command validation.
- Hysteresis, heater runtime limits, sensor disconnection, and fail-safe behavior.
- Simulator/real-device source labels and device reconnection.

13. Known Limitations and Open Decisions

- The DHT11 may not provide adequate accuracy for sustained high-humidity cultivation; evaluate it or upgrade to a suitable sensor.
- The moisture module must be identified and calibrated for the actual substrate.
- The humidifier button circuit must be inspected before designing electronic button automation.
- A fan can circulate or exchange air but cannot cool below ambient temperature without an appropriate cooling mechanism.
- The correct cultivation targets depend on species, strain, substrate, and stage.
- Yield and harvest predictions require ground-truth observations across growing cycles.
- AI provider endpoint/model details must be verified before integration.

14. Final Project Outcome

MYCO AI should mature from a working environmental monitor into a reliable cultivation data platform. Its value will come from collecting trustworthy sensor readings, recording actual mushroom development, providing safe environmental control, and validating predictions against real outcomes—not merely from displaying AI-generated text.

Reference status: This document is a project concept and implementation guide, not a validated cultivation protocol or finished electrical design. Verify all electrical ratings, safety measures, sensor behavior, and species-specific growing conditions before operating the

physical system.